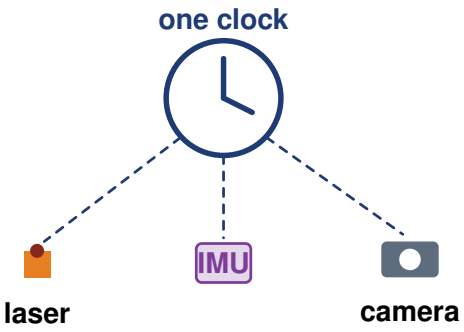
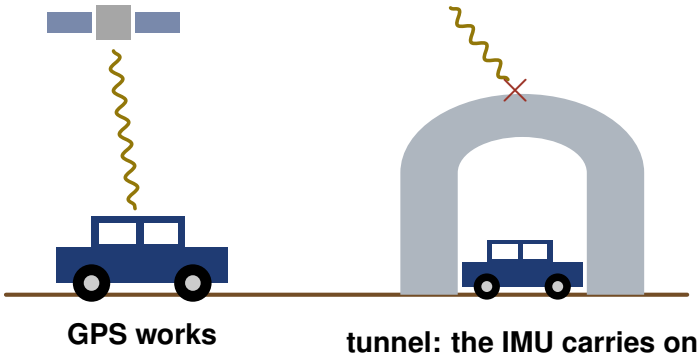
Timing, GNSS and Motion Compensation

BIG IDEA

Every sensor must share one clock, and GPS and the IMU make a perfect team.





THE IDEA

If your camera's clock is one second slow, your photo of a runner crossing the finish line gets labeled at the wrong time, and the runner seems to be in the wrong place. Every sensor on a mapping car must use the *same* clock, set perfectly, so every laser dot is matched to exactly where the car was at that instant.

REMEMBER

- ✓ One clock for the laser, the IMU, the camera and GPS.
- ✓ GPS never drifts but can disappear. The IMU never stops but drifts. Together they are great.
- ✓ The car moves while the laser spins, so each dot is moved back to where it really was (**de-skewing**).

TRY THIS

Take a photo of something moving fast, like a ball being thrown. Is it blurry? The LiDAR has the same problem and fixes it by using the time of every dot.

The full module is in the Reference Book.